

Technical Documentation: IPv6 Dual-Stack Deployment via WireGuard

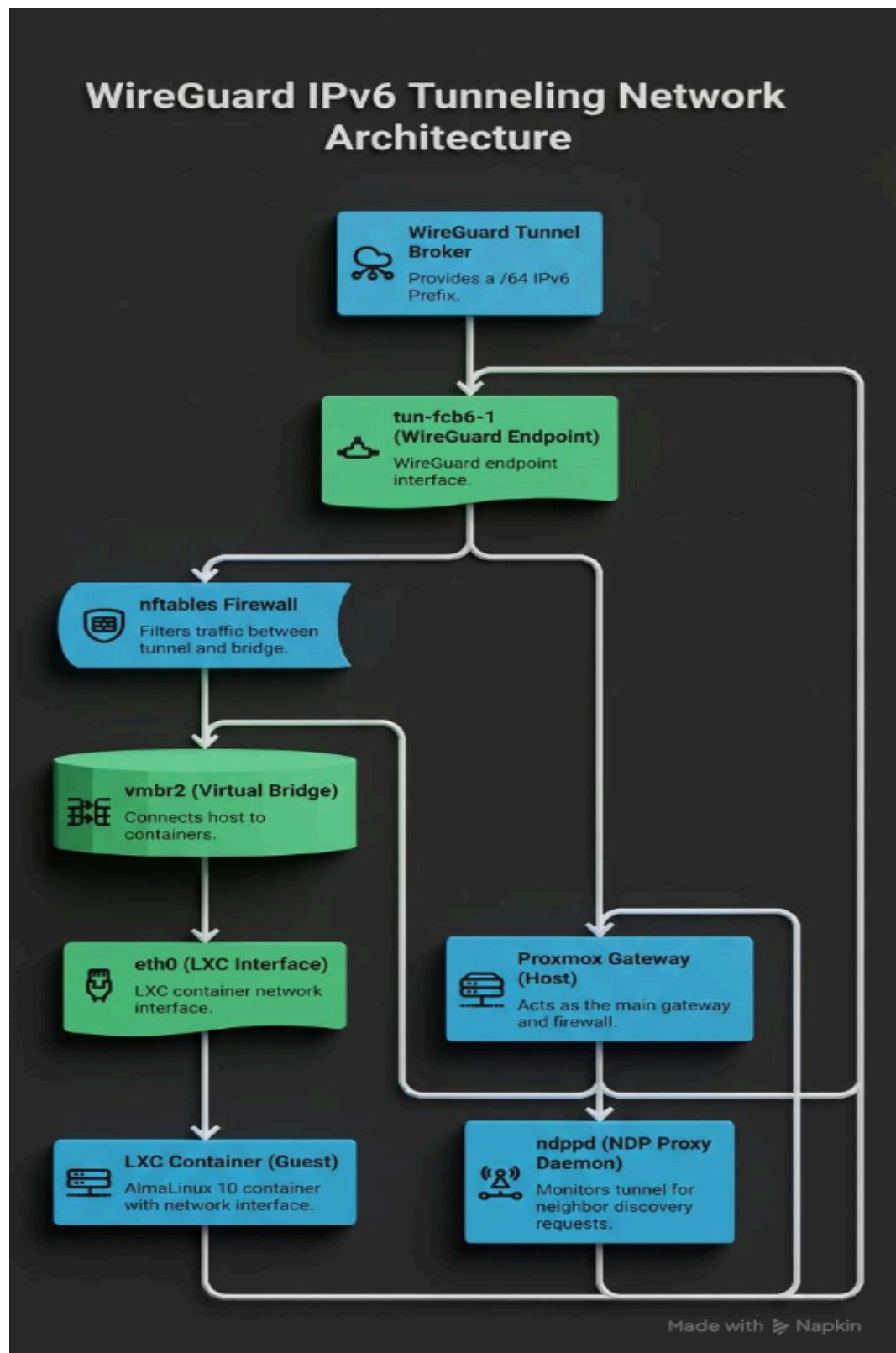
Project Scope: Implementing public IPv6 connectivity for isolated LXC containers.

Date: April 2026

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1. Project Objective & Operational Proof

The primary goal was to provide native IPv6 connectivity to isolated LXC containers (AlmaLinux 10) within a Proxmox environment, bypassing the limitation of no native IPv6 support from the Internet Service Provider (ISP).



1.1 Connectivity Verification (The "Smoking Gun")

The ultimate proof of a successful Dual-Stack implementation is the ability to resolve and route traffic over both protocols simultaneously.

```
Linux) on node raspurin No tags
[root@AlmaLinux ~]# ping -c 4 google.com
PING google.com (142.251.14.101) 56(84) bytes of data.
64 bytes from pm-in-f101.1e100.net (142.251.14.101): icmp_seq=1 ttl=113 time=62.2 ms
64 bytes from pm-in-f101.1e100.net (142.251.14.101): icmp_seq=2 ttl=113 time=62.7 ms
64 bytes from pm-in-f101.1e100.net (142.251.14.101): icmp_seq=3 ttl=113 time=61.8 ms
64 bytes from pm-in-f101.1e100.net (142.251.14.101): icmp_seq=4 ttl=113 time=62.6 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 61.797/62.302/62.688/0.348 ms
[root@AlmaLinux ~]# curl -6 https://ifconfig.co
2a03:94e0: [redacted]::2
[root@AlmaLinux ~]#
```

AlmaLinux terminal showing: `ping -c 4 google.com` (IPv4 success) `curl -6 https://ifconfig.co` (IPv6 success showing your assigned prefix)

Purpose: Demonstrates that the entire stack (Routing, DNS, MTU) is perfectly tuned.

2. Infrastructure Layer: WireGuard Tunneling

A tunnel-based architecture was implemented using the WireGuard protocol, operating in an L3 Tunneling model.

2.1 Tunnel Persistence and Stability

The backbone of this solution is a stable WireGuard tunnel. Monitoring handshakes is crucial for verifying link health.

```
persistent keepalive: every 25 seconds

interface: tun-fcb6-1
  public key: 6POCKfw8+nlHupyJ8g5s8Bu [redacted] A+ect9yxF2k=
  private key: (hidden)
  listening port: 39467
  fwmark: 0xca6c

peer: 3uMqzEwOBHoDS57yg3MAtgMUvhZa0 [redacted] +XwwxUU=
  endpoint: 185.243. [redacted]:51820
  allowed ips: ::/0
  latest handshake: 36 seconds ago
  transfer: 570.47 KiB received, 405.19 KiB sent
  persistent keepalive: every 25 seconds
root@rasputin:~#
```

Proxmox terminal showing: `wg show`

Purpose: Highlights the "latest handshake" and "transfer" metrics, proving the tunnel is active and stable.

3. Security Layer: Modern nftables Firewall

A modern **nftables** firewall was deployed to replace the legacy iptables framework. This move ensures high-performance packet filtering and stateful inspection.

3.1 Security Ruleset

A "Default Drop" policy is applied to all incoming traffic on the tunnel interface to ensure **Cybersecurity** best practices.

```
root@rasputin:~# nft list ruleset
table ip6 filter v6 {
    chain input {
        type filter hook input priority filter; policy accept;
        iifname "tun-fcb6-1" ct state established,related accept
        iifname "tun-fcb6-1" meta l4proto ipv6-icmp accept
        iifname "tun-fcb6-1" drop
    }

    chain forward {
        type filter hook forward priority filter; policy accept;
        iifname "tun-fcb6-1" ct state established,related accept
        oifname "tun-fcb6-1" accept
    }
}
table ip6 wg-quick-tun-fcb6-1 {
    chain preraw {
        type filter hook prerouting priority raw; policy accept;
        iifname != "tun-fcb6-1" ip6 daddr 2a03:94e0: [REDACTED]::1 fib saddr type != local drop
        iifname != "tun-fcb6-1" ip6 daddr 2a12:bec0: [REDACTED]:1 fib saddr type != local drop
        iifname != "tun-fcb6-1" ip6 daddr 2a05:1083: [REDACTED]:1 fib saddr type != local drop
        iifname != "tun-fcb6-1" ip6 daddr fde4:5a50: [REDACTED]:2 fib saddr type != local drop
    }

    chain premangle {
        type filter hook prerouting priority mangle; policy accept;
        meta l4proto udp meta mark set ct mark
    }

    chain postmangle {
        type filter hook postrouting priority mangle; policy accept;
        meta l4proto udp meta mark 0x0000ca6c ct mark set meta mark
    }
}
root@rasputin:~#
```

Proxmox terminal showing: `nft list ruleset`

Purpose: Shows your `filter_v6` table and `forward` chains. This proves you are not just "making it work," but "making it secure."

4. Debugging & L3 Infrastructure Logic

During deployment, critical issues such as **Routing Loops** and **NDP Failures** were identified and mitigated through advanced L3 configuration.

4.1 Routing & Neighbor Discovery (NDP) Proxying

To ensure external nodes could resolve the container's address and to eliminate routing loops, a specific host route and NDP Proxying were implemented.

```
root@rasputin:~# ip -6 route show dev vmbr2 && ip -6 neigh show proxy
2a03:94e0: [REDACTED]::2 metric 1024 pref medium
2a03:94e0: [REDACTED]::/64 proto kernel metric 256 pref medium
fe80::/64 proto kernel metric 256 pref medium
2a03:94e0: [REDACTED]::2 dev tun-fcb6-1 proxy
root@rasputin:~#
```

Command: `ip -6 route show dev vmbr2 && ip -6 neigh show proxy`

Purpose & Logic:

- **Host Route:** The routing table shows a **specific /128 route** (`::2 dev vmbr2`). By applying **Longest Prefix Match (LPM)** logic, this route takes precedence over the general tunnel route, successfully breaking the routing loop (Hop Limit Exceeded).
- **NDP Proxy:** The `proxy` status confirms that the Proxmox host is intercepting Neighbor Discovery requests from the WireGuard tunnel and answering on behalf of the container, making the LXC visible to the external network

5. Guest Optimization (AlmaLinux 10)

Persistence ensures that the configuration survives reboots. Proper MTU and DNS settings at the guest level prevent packet fragmentation.

5.1 System Persistence via NetworkManager

The IPv6 stack within the container was optimized for the WireGuard encapsulation overhead.

```
[root@AlmaLinux ~]# nmcli connection show eth0 | grep -E 'mtu|ipv6.dns|ipv6.method'
802-3-ethernet.mtu: 1400
ipv6.method: manual
ipv6.dns: 2606:4700:4700::1111,2606:4700:4700::1001
ipv6.dns-search: --
ipv6.dns-options: --
ipv6.dns-priority: 0
ipv6.mtu: auto
[root@AlmaLinux ~]#
```

AlmaLinux terminal showing: nmcli connection show eth0

Purpose: Highlights the **MTU 1400** setting and specific IPv6 DNS resolvers. It shows attention to detail in system optimization.

6. Debugging & Problem Solving (Key Competencies)

A successful deployment in a complex networking environment requires more than just configuration; it requires the ability to diagnose and mitigate protocol-level conflicts.

6.1 Routing Loop (Hop Limit Exceeded)

- **Issue:** Upon initial setup, internal traffic was caught in an infinite loop, causing "Hop Limit Exceeded" errors.
- **Root Cause:** The host was attempting to route packets destined for the container (: :2) back through the default tunnel route (: :/0).
- **Resolution:** Implemented a **Host Route** with a /128 prefix on the Proxmox bridge (vmbbr2). By leveraging the **Longest Prefix Match (LPM)** algorithm, the kernel prioritized the local interface over the tunnel, breaking the loop.

6.2 Neighbor Discovery Protocol (NDP) Failure

- **Issue:** The AlmaLinux container remained unreachable from external nodes despite valid addressing.
- **Root Cause:** WireGuard is a Layer 3 protocol and does not natively pass Layer 2 NDP packets. The tunnel broker could not resolve the Link-Layer address of the container.
- **Resolution:** Deployed and configured **ndppd** (NDP Proxy Daemon). This allowed the Proxmox host to act as a proxy, intercepting and answering Neighbor Solicitation requests on behalf of the LXC container.

6.3 MTU Mismatch & TCP Handshake Hangs

- **Issue:** ICMP traffic worked, but encrypted HTTPS/TLS sessions would hang during the handshake.
- **Root Cause:** WireGuard's 60-byte encapsulation overhead reduced the effective payload size, causing large TLS certificates to be fragmented or dropped.
- **Resolution:** Optimized the **Maximum Transmission Unit (MTU) to 1400 bytes** across all interfaces. This ensures full compatibility with the tunnel overhead and prevents application-layer timeouts.

7. Conclusion & Project Impact

This project successfully delivered a fully functional **IPv6 Dual-Stack environment** in a restricted ISP infrastructure.

Key Achievements:

- **Scalability:** The architecture allows for additional LXC containers to be deployed with unique public IPv6 addresses within the **/64** prefix.
- **Security-First Design:** Using **nftables** instead of legacy tools ensures a modern, high-performance security posture.
- **Operational Excellence:** The final configuration is persistent, surviving reboots and maintaining high availability for the AlmaLinux services.

The implementation proves that with advanced knowledge of **L3 Routing** and **Tunneling Protocols**, it is possible